

Nearly Linear Time Algorithm for Follow-the-Perturbed-Leader with Best-of-Both-Worlds Guarantee in Bandit Problems

Master's Thesis Defense

Systems Science Course

Graduate School of Informatics, Kyoto University

Botao Chen (陳 伯韜)

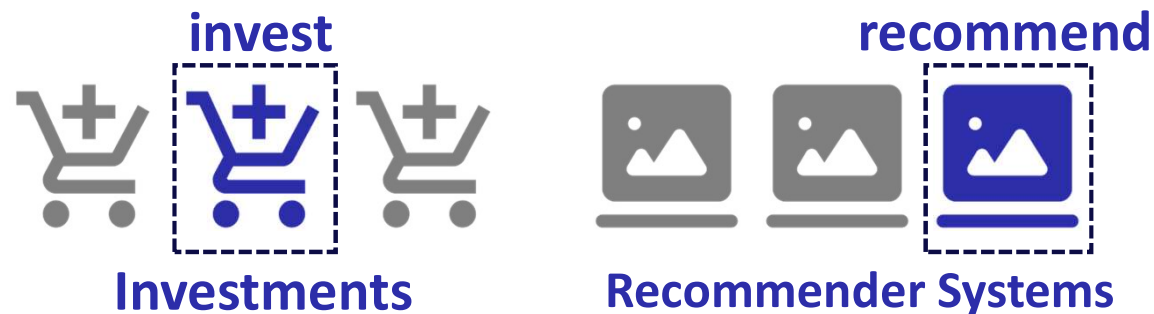
2026/2/3

Multi-armed Bandit Problems

- Model of a gambler playing a slot machine with K arms.
- He pulls one arm $I_t \in \{1, 2, \dots, K\}$ at each round.
- Once arm i is pulled, he will observe the loss from arm i .
- Loss cannot be observed unless the arm is pulled.
- Considering the loss $\ell_{t,i} \in [0, 1]$, the goal is to minimize the regret

$$\text{Regret}(T) = \max_{i \in [K]} \mathbb{E} \left[\sum_{t=1}^T (\ell_{t,I_t} - \ell_{t,i}) \right].$$

Application:



Two Settings of Environments

There are two settings of environments generating the losses:

Stochastic Setting

- The loss $\ell_{t,i}$ from arm i in round t is distributed by $\mathcal{D}_i$ with mean μ_i .
- The optimal arm is $i^* = \arg \min_{i \in [K]} \mu_i$.

Adversarial Setting

- The loss of each arm at each round $\{\ell_t\}_{t=1}^T$ is chosen **adversarially**.
- The optimal arm set is $i^* = \arg \min_{i \in [K]} \mathbb{E} \left[\sum_{t=1}^T \ell_{t,i} \right]$.

Follow-The-Regularized-Leader (FTRL)

Best-of-Both-Worlds (working well in both sto. and adv. settings):

- Achieve $O(\sqrt{KT})$ regret in adversarial setting.
- Achieve $O(\sum_{i \neq i^*} \frac{\log T}{\Delta_i})$ regret in stochastic setting.
gap between μ_i and the smallest expected loss μ_{i^*}

Follow-The-Regularized-Leader (FTRL)

- A policy choosing an arm according to the arm-selection probability $w_t \in \mathcal{P} = \{w \in (0, 1)^K : \sum_{i=1}^K w_i = 1\}$ given by

$$w_t = \arg \min_{w \in \mathcal{P}} \left\{ \underbrace{w^\top \hat{L}_t}_{\text{pull the arm with small cumulative loss}} + \underbrace{\varphi_t(w)}_{\text{penalty for biased selection probability}} \right\}.$$

- $\hat{L}_t = \sum_{s=1}^{t-1} \hat{\ell}_s \in \mathbb{R}^K$: unbiased estimator of the loss vector.
- $\varphi_t(w) = \varphi(w)/\eta_t$: regularization term for learning rate $\eta_t > 0$.
- FTRL can achieve **Best-of-Both-Worlds** guarantee. [Tsallis-INF, Zimmert+2021]
- Needs to **solve an optimization problem**, leading to **high computational cost**.

Follow-the-Perturbed-Leader (FTPL)

Follow-The-Perturbed-Leader (FTPL) [Neu+2016]

- At each round t , FTPL chooses an arm $I_t \in [K] = \{1, 2, \dots, K\}$ given by

$$I_t = \arg \min_{i \in [K]} \left\{ a^T (\eta_t \hat{L}_t - r_t) \right\}$$

where $r_t = (r_{t,1}, \dots, r_{t,K})$ is perturbation sampled from some distribution.

- FTPL with **Fréchet-type perturbation** achieves **Best-of-Both-Worlds** [Honda+2023]
- No** need to **solve an optimization problem**.
- Construct loss estimators $\hat{\ell}_t$ by **Geometric Resampling**. (*mentioned later*)

Example (○ : selected):

	Arm 1	Arm 2	Arm 3	Arm 4	Arm 5
$\eta_t \hat{L}_t =$	2,	2,	1,	4,	5
perturbation value $r_t =$	4,	5,	2,	3,	3
$\eta_t \hat{L}_t - r_t =$	-2,	○ -3,	-1,	1,	2

Perturbation Distributions

FTPL with any of the following **perturbation distribution** achieves Best-of-Both-Worlds in standard bandits (m=1 case):

α -type Fréchet distribution ($\alpha > 1$)

- The CDF is expressed as

$$F(x) = e^{-1/x^\alpha}, x \geq 0.$$

α -type Pareto distribution ($\alpha > 1$)

- The CDF is expressed as

$$F(x) = 1 - x^{-\alpha}, x \geq 1.$$

Therefore, in this study (m $\geq$ 1 case) as an extension of standard bandits, we still consider the above both distributions.

Partial Feedback: Have to Construct Unbiased Loss Estimator

Partial Feedback (Bandit Feedback)

In bandit problems, only the loss ℓ_{t,l_t} from the chosen arm l_t can be observed



We have to construct an estimator $\hat{\ell}_t \in [0, \infty)^K$ s.t. $\mathbb{E}[\hat{\ell}_t] = \ell_t$



Importance-Weighted (IW) Estimator: $\hat{\ell}_{t,i} = \mathbb{1}(l_t = i)\ell_{t,i} w_{t,i}^{-1}$ — $w_{t,i}$: Probability that choose arm i

IW Estimator:

- Widely used by FT~~R~~L, which explicitly computes
- FTPL doesn't need $w_{t,i}$ for arm-selection, and it is very expensive to compute

Solution: Geometric Resampling [Neu+2016] ↓ to give unbiased estimator $\widehat{w_{t,i}^{-1}}$ for $w_{t,i}^{-1}$

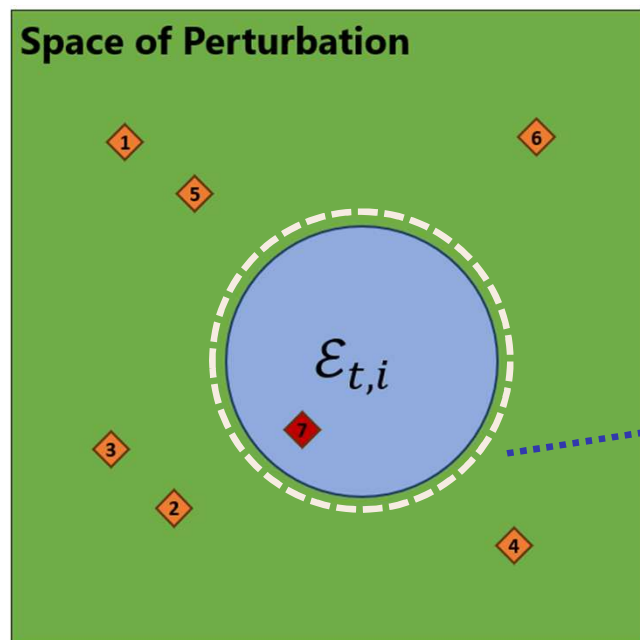
Loss Estimator: $\hat{\ell}_{t,i} = \mathbb{1}(l_t = i)\ell_{t,i} \widehat{w_{t,i}^{-1}}$

Loss estimator is still unbiased!

Geometric Resampling (GR): Construction of Loss Estimators

To construct:

Loss Estimator: $\hat{\ell}_{t,i} = \mathbb{1}(I_t = i) \ell_{t,i} \widehat{w}_{t,i}^{-1}$



Steps of giving $\widehat{w}_{t,i}^{-1}$

Sample r'_t from perturbation distribution

JUDGE

$$r'_t \in \mathcal{E}_{t,i} = \left\{ r'_t : I_t = \arg \min_{i \in [K]} \left\{ \eta_t \hat{L}_{t,i} - r'_{t,i} \right\} \right\}?$$

YES

Set $\widehat{w}_{t,i}^{-1} = (\text{Number of Resampling})$

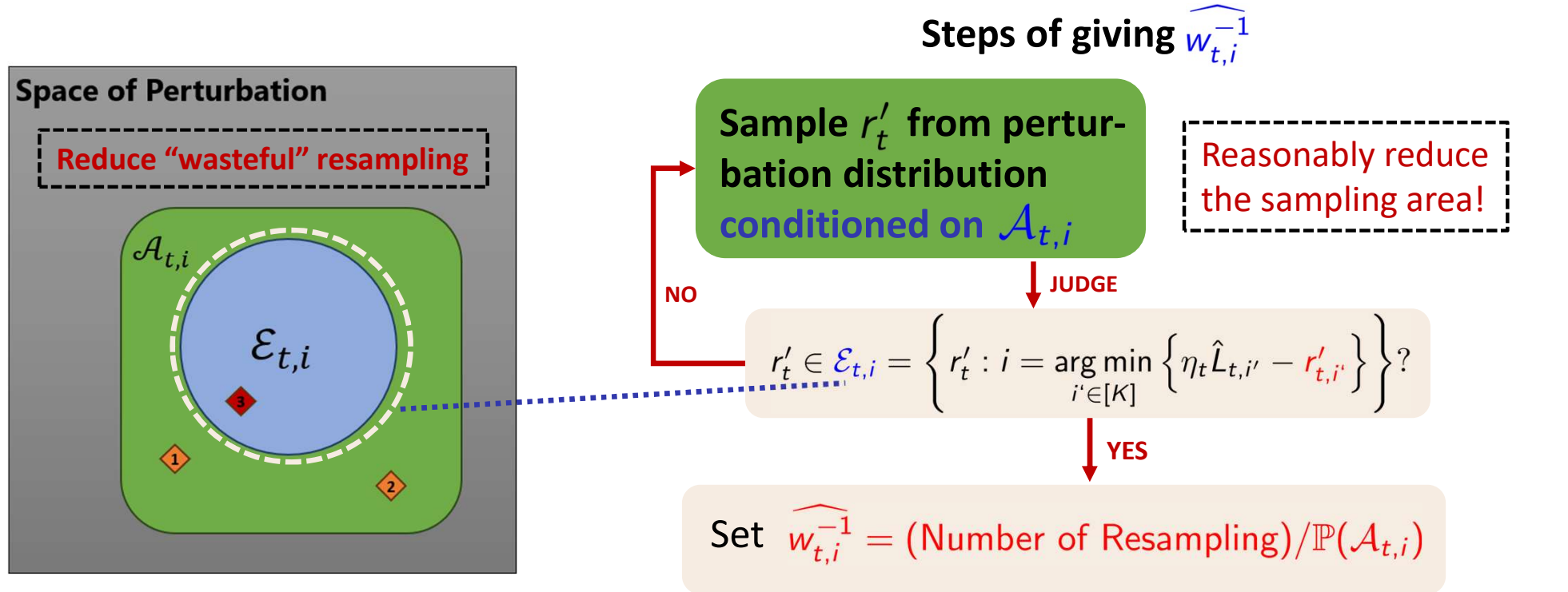
Geometric Resampling

$\widehat{w}_{t,i}^{-1}$ is unbiased since the number of resampling follows geometric distribution with successful probability $\mathbb{P}(r'_t \in \mathcal{E}_{t,i}) = 1/w_{t,i}$ so that

$$\mathbb{E}[(\text{Number of Resampling})] = \frac{1}{\mathbb{P}(r'_t \in \mathcal{E}_{t,i})} = \frac{1}{w_{t,i}}$$

However, the average complexity of GR is $O(K^2)$, which is worse than FTRL (empirically $O(K)$).

Conditional Geometric Resampling (CGR): Reduce Complexity



Conditional Geometric Resampling

The number of resampling follows geometric distribution with successful probability $\mathbb{P}(r'_t \in \mathcal{E}_{t,i} | \mathcal{A}_{t,i})$ so that

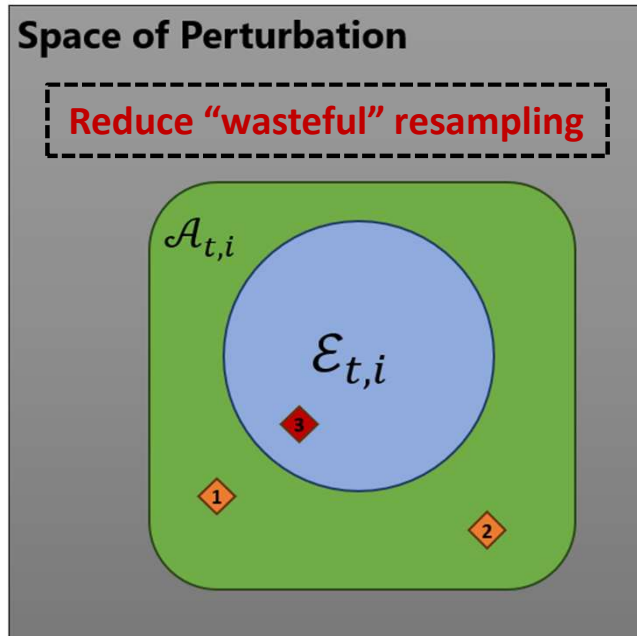
$$\mathbb{E}[(\text{Number of Resampling})] = \frac{\mathbb{P}(\mathcal{A}_{t,i})}{\mathbb{P}(r'_t \in \mathcal{E}_{t,i})} = \frac{\mathbb{P}(\mathcal{A}_{t,i})}{w_{t,i}}$$

Therefore,

$$\mathbb{E}[\widehat{w_{t,i}^{-1}}] = \frac{\mathbb{P}(\mathcal{A}_{t,i})}{w_{t,i}} / \mathbb{P}(\mathcal{A}_{t,i}) = \frac{1}{w_{t,i}}$$

Keep unbiased!

Design of Condition $\mathcal{A}_{t,i}$



Conditional Geometric Resampling

$\mathcal{A}_{t,i}$ should satisfy the following requirements:

1. $\mathcal{A}_{t,i}$ is necessary condition of $\mathcal{E}_{t,i}$, i.e. $\mathcal{E}_{t,i} \subset \mathcal{A}_{t,i}$

$$r'_t : i = \arg \min_{i' \in [K]} \left\{ \eta_t \hat{L}_{t,i'} - r'_{t,i'} \right\}$$

2. The sampling method of r'_t conditioned on $\mathcal{A}_{t,i}$ can be efficient
3. $\mathbb{P}(\mathcal{A}_{t,i})$ can be computed easily



Type-I: $\mathcal{A}_{t,i} := \left\{ r'_t : r'_{t,i} = \max_{i' : \hat{L}_{t,i'} \leq \hat{L}_{t,i}} r'_{t,i'} \right\}$

Type-II:

$$\mathcal{A}_{t,i} := \text{Type-I} \cap \left\{ r'_t : r'_{t,i} \geq \eta_t (\hat{L}_{t,i} - \min_{i' \in [K]} \hat{L}_{t,i'}) \right\}$$

Properties of CGR I and CGR II

CGR Type-I: $\mathcal{A}_{t,i} := \left\{ r'_t : r'_{t,i} = \max_{i': \hat{L}_{t,i'} \leq \hat{L}_{t,i}} r'_{t,i'} \right\}$

- Average complexity: $O(K \log K)$. **K: the number of arms**
- Sampling method is very efficient:
STEP 1: Sample r'_t i.i.d. from the perturbation distribution (the same as original GR)
STEP 2: Swap the value of components s.t. r'_t satisfies $\mathcal{A}_{t,i}$ (**with almost no additional computational cost**)

CGR Type-II: $\mathcal{A}_{t,i} := \text{Type-I} \cap \left\{ r'_t : r'_{t,i} \geq \eta_t (\hat{L}_{t,i} - \min_{i' \in [K]} \hat{L}_{t,i'}) \right\}$

- Average complexity: $O(K \log K)$.
- The number of resampling is further reduced compared with CGR I.
- Can terminate in advance when the number of resampling reaches $\lceil (K \vee 4) \log t \rceil$ without sacrificing the BOBW guarantees.
- The sampling method is a bit time-consuming compared with CGR I.

Experiment 1: Reduced Variance by CGR Improves Regret

Experiment: Regret performance with the number of arms $K = 8$

We compare FTPL with the following BOBW policies

- T-INF IW/RV: FTRL-based policy with IW/RV estimator
- TS-Prod: Policy with theoretical guarantee but empirically performs badly.

Regret (lower is better): $\boxed{\text{FTPL CGR II} \leq \text{FTPL CGR I} \leq \text{FTPL GR}}$

Thanks to the reduced variance of $\widehat{w_{t,i}^{-1}}$

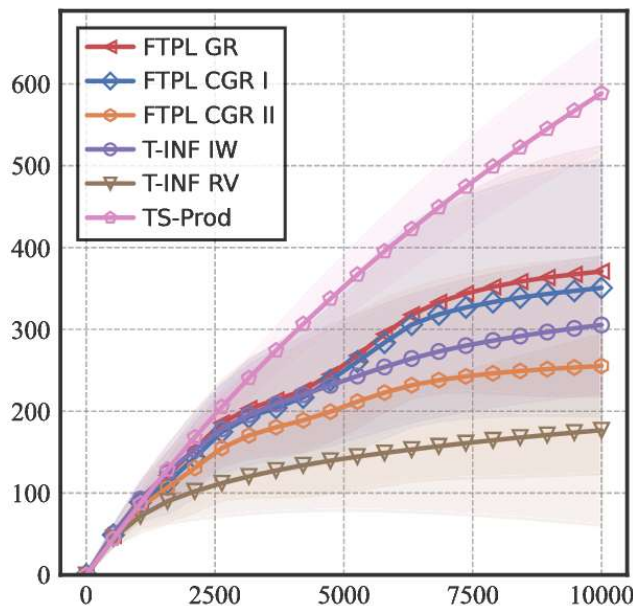


Figure 1: Adversarial settings.

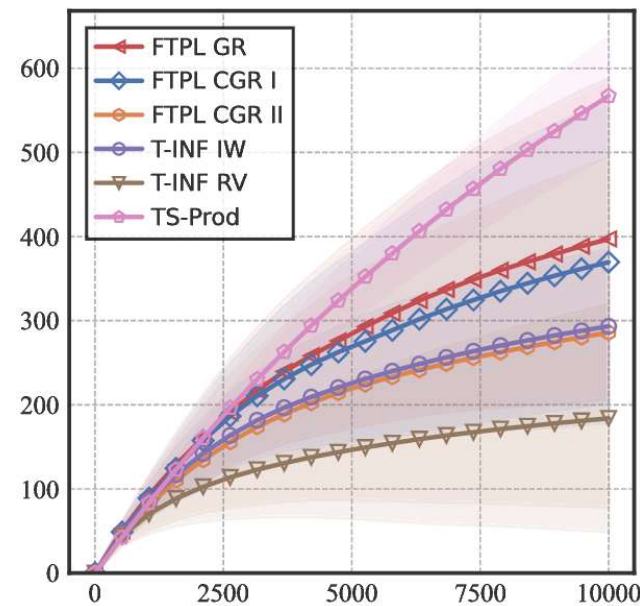


Figure 2: Stochastic settings.

Experiment 2: CGR II Samples Less but CGR I Runs Faster

Experiment: Expected number of resampling and runtime for adversarial setting

Expected number of resampling: FTPL CGR II $\leq$ FTPL CGR I $\leq$ FTPL GR

Runtime:

- As K increases, FTPL GR grows sharply.
- CGR I runs faster than CGR II, thanks to the efficient sampling method of CGR I.

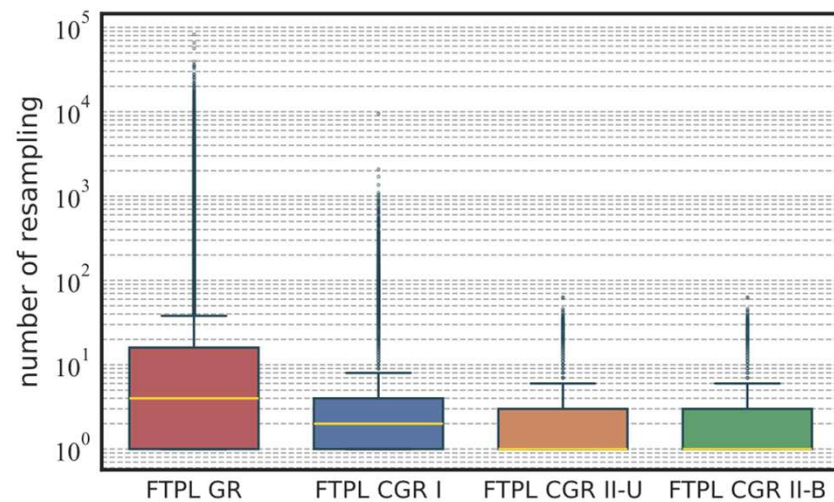


Figure 3: Expected number of resampling.

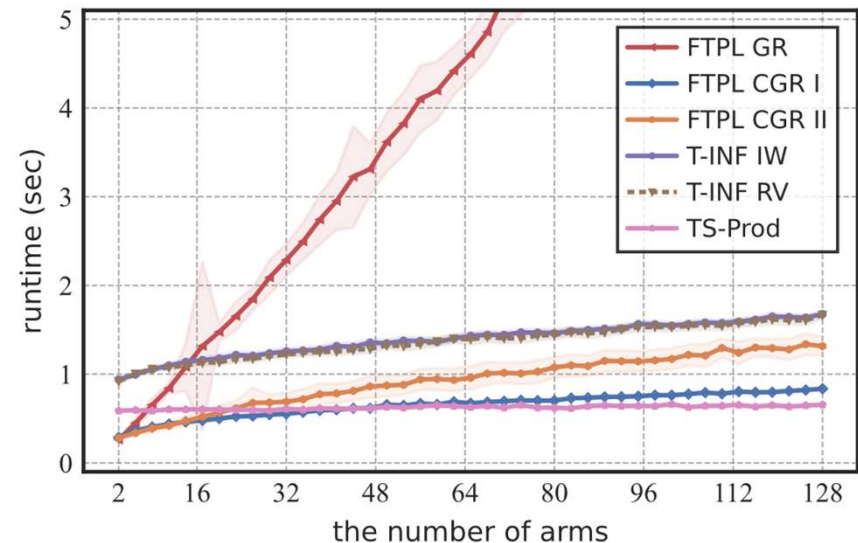


Figure 4: Runtime (sec).

We studied the complexity of FTPL:

- The computational complexity of GR is $O(K^2)$, which causes inefficiency for large number of arms K .
- We proposed a novel technique called Conditional Geometric Resampling (CGR I, CGR II), which reduces the complexity of FTPL to $O(K \log K)$ without sacrificing the regret guarantees.
- We confirm through experiments that CGR does not only significantly **improve the runtime** but also **achieve better regret** thanks to the stable loss estimation.